THE PROBLEM · WHICH ONE IS THE BOT?

One by one, a bot and a human now read the same

Two comments under the same war-news post. Pick the bot.

Comment A · 🤖 Bot

"Honestly can't believe they left the town undefended again. Third night this week and nobody answers for it."

Comment B · 🙂 Human

"Same here. Three nights this week and still nobody is accountable. I'm tired of it."

Illustrative pair written for this slide, not taken from the data.

THE PROBLEM · FLIP THEM

Swap the labels. Nothing in the text changes.

So we stop judging single comments. A swarm is different in three ways.

Comment A · 🙂 Human

"Honestly can't believe they left the town undefended again. Third night this week and nobody answers for it."

Comment B · 🤖 Bot

"Same here. Three nights this week and still nobody is accountable. I'm tired of it."

Coordinated

Many accounts act together, within minutes.

Directed

Aimed at the same posts and channels.

Intentional

Repeated again and again, not by chance.

OUR METHOD · TIMING + MEANING

Timing alone is not enough. Timing plus meaning exposes the hidden pattern.

Same time
8.4M pairs within 5 min

→

Same meaning
62,499 also say the same thing

→

Again and again
822 pairs repeat on 2+ days

→

Beats chance?
keep only what luck cannot explain

8.4M
pairs that are close in time

→

62,499
also close in meaning

→

822
repeat on 2+ days

Timing signal Meaning signal (embedding space) Check against shifted timelines

Source: swarm-detector analysis, 15 Telegram channels, 14–21 Oct 2026, 300 s window. Reply chats are removed first. Validation on a separate test set comes next.